- Fixed implementation with pre-defined constraints
- Limited availability of number of hard IPs with respect to particular FPGAs

Firm IP cores. Firm IP cores are also known as semi-hard IP cores. These are a form of gate-level netlist, where you have the flexibility to place the module in the FPGA as per usage and with minimal user-programmable configurations.

For example, if a third-party IP is targeted at Xilinx FPGA, then the IP provided will be .ngc file. You can integrate this file with your project and instantiate it as a component in the top level to interconnect with other modules, and then proceed with synthesis.

To use a firm IP core, you should have proper FPGA resource planning and requirement specifications before procuring the IP for the project.

Xilinx Coregen-generated IP cores (like FIFO, shift registers and memory interface cores) can be grouped into the firm IP core category. You have to include .ngc/.xco in the project directory (for Xilinx), and specify the instantiation in the top file. Instantiated components can be moved around within the FPGA to meet performance and timing.

Advantages are:

- Modifications allowed to some extent
- Functionality and performance are measurable
- Resource utilisation considers firm IP logic area
- Completely tested
- Documentation may be available up to some level

Disadvantages are:

- Limited portability
- Modifications to source not possible
- May be licensed based on cost
- Source utilisation matters while considering the FPGA
- Timing/performance may have impact

Soft IP cores. These are completely flexible and do not depend on vendor technology. These can be ported across various FPGA platforms. The IPs are developed using HDL languages and you are provided with source codes, so the IPs can be modified according to your application and easily integrated with your modules. These are reusable and can be targeted at many variants of FPGAs.

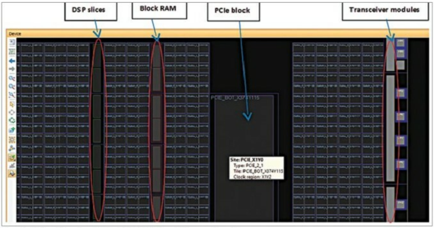

Fig. 2: Xilinx Vivado - Virtex-7 FPGA hard IP core locations

In Xilinx FPGAs, ARM, Zynq and PowerPC processors fall under hard IP category, whereas Microblaze falls in the soft IP group.

Similarly, in Altera, ARM, Intel ATOM processors come under hard IP core, and Nios-II processor under soft IP core.

Examples for other soft IP cores are 8051 microcontrollers IP, I2C controllers, SPI controllers, standard bus interfaces or any open core whose source can be modified.

These modules should be used as primitive instantiation in the custom RTL so that these get connected to user logic. In some cases, DDR2/3 cores are also specific to some banks of FPGAs.

Advantages are:

- Portable across FPGAs, independent of target technology/vendor
- Available as open source IP
- Easy to modify and implement as per custom applications
- Complete data sheet/documentation available for originally-targeted device/technology

Disadvantages are:

- IP cost may be high when compared to firm IP core since source code provided by third party
- Individual IP core licence may be required
- Performance/timing may vary with originally-targeted device/technology
- Documentation might be generic or specific to originally-targeted device/technology
- Support may not be available (or limited support available) for custom modifications
- Extra effort required while targeting custom technology/device
- In-depth understanding of design required before modifications can be made

Several factors must be considered while selecting the suitable IP such as cost of IP, effort required for customisation (for soft IP core), time to market, FPGA family, availability of simulation environment (unit and integrated levels) of IP, documentation, board-level evaluation of IP and support during development, among others. Nowadays, since most basic IP cores come with FPGA implementation tools for free, it makes time-to-market faster and results in lower costs.

In this article, we have taken Xilinx and Altera as examples. But you can find other vendors as per your FPGA designs. So depending on complexity, timeline and overall project cost, you can choose proper IP core type. EFY